Avery Yang

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Education

Cornell University, College of Arts and Sciences **Ithaca, NY**
Bachelor of Arts in Computer Science and Cognitive Science *Expected May 2028*
Relevant Coursework: Computer System Organization, Data Structures and Functional Programming, Discrete Structures, Object-Oriented Programming & Data Structures, Intro to Data Science, Trends in Web Dev, Linear Algebra

Skills

Languages: Python, Java, JavaScript/TypeScript, C, C++, SQL, OCaml, HTML/CSS, Linux/Bash
Frameworks & Libraries: PyTorch, LangChain, scikit-learn, spaCy, FastAPI, React, Express.js, Node.js, pandas, NumPy
Tools & Infra: Docker, Prisma, Render, Vercel, Firebase, PostgreSQL, Postman, Git/GitHub, Figma

Professional Experience

Computational Linguistics Lab, Cornell University **Ithaca, NY**
Research Assistant *May 2026-Present*
- Developing an ASL sign recognition system in React, PyTorch, and FastAPI that lets learners identify partially-remembered signs from live video, replacing manual video search with ranked matches and demonstrations
- Implementing a vector-based keyframe extraction algorithm using motion-threshold filtering and farthest-point sampling with MediaPipe to isolate hand and body landmarks for sign recognition
- Designed and deployed REST APIs on a Cornell Apache server to process video inputs and serve recognition results
- Training and integrating PyTorch classification models on the ASL-LEX and SEM-LEX datasets, spanning ~85k video samples, for sign recognition and ASL phonological parameter identification

Hack4Impact **Ithaca, NY**
Software Developer *November 2025-Present*
Medic | *Community Health Toolkit Impact Dashboard*
- Building REST endpoints in TypeScript, Express, and PostgreSQL to serve connected-user counts, CHT version distribution, and deployment monitoring data for CHT deployments across 24 countries and 180k+ health workers
- Linking monitoring snapshots with deployment records, preserving measurement windows and data-quality indicators
- Implementing server-side sharing controls and API tests for restricted deployments, stale snapshots, and failed metrics

Anabel's Grocery | *Volunteer Shift Management*
- Shipped 10+ REST API endpoints for shifts and swap requests, supporting scheduling coordination for 50+ volunteers
- Integrated Firebase Admin SDK for server-side token verification, securing volunteer and admin authentication

NeuPals Holdings - Cricket Media **Fairfax, VA**
Artificial Intelligence Product Research Intern *July 2025-August 2025*
- Conducted competitive market analysis and prototyped AI feature concepts in Figma to demonstrate opportunities for an edtech platform serving international students

Projects

Readability Lab | *React, JavaScript, Python, FastAPI, LangChain, spaCy, Gemini, Vercel, Render* *Sept 2026*
- Built and deployed a full-stack text-adjustment tool in React and FastAPI that bidirectionally rewrites passages toward a user-selected reading level, visualizing readability and syntactic complexity changes
- Developed a LangChain verify-and-revise pipeline that evaluates each LLM rewrite, supplies metric-based revision feedback, and returns the closest result within three iterations
- Integrated spaCy and textstat to quantify readability and syntactic complexity, achieving grade-level shifts from 15.7 to 4.5 and 0.6 to 9.6 in representative tests

Film Diary | *React, TypeScript, Node.js, Express.js, Docker, Vercel* *Jan 2026-May 2026*
- Independently built and deployed a full-stack movie discovery and logging web application, containerized with Docker and hosted on Vercel
- Integrated the TMDB API to support live search with movie metadata, posters, and popularity ranking feeds
- Built CRUD functionality for user reviews and watchlist management with Firebase auth and per-user data persistence

Donut Shop Simulation Game | *OCaml* *Jan 2026-May 2026*
- Implemented core gameplay systems in a Model-View-Controller OCaml codebase (2,400+ lines), including functional game-state transitions, player interaction logic, collision detection, and station-specific timers
- Wrote unit tests for timing and gameplay logic and implemented a replay feature to support debugging and validation